

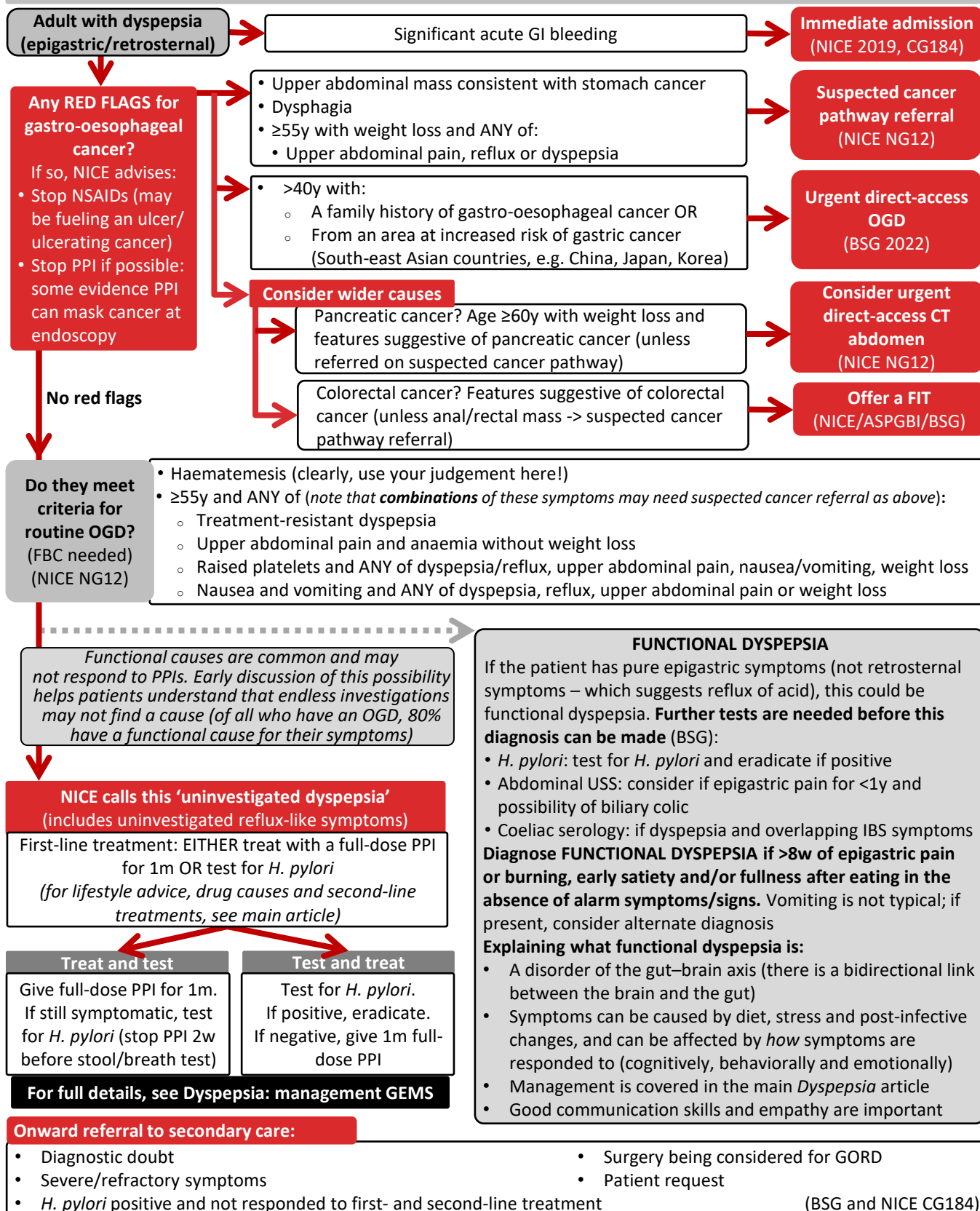
# Dyspepsia in adults: framework

(Gut 2022;71:1697, Gut 2022;71:1939, NICE 2025, NG12, NICE 2019, CG184)



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**Biliary colic and cardiac disease can mimic dyspepsia – if suspected, investigate**



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# Dyspepsia in adults: management

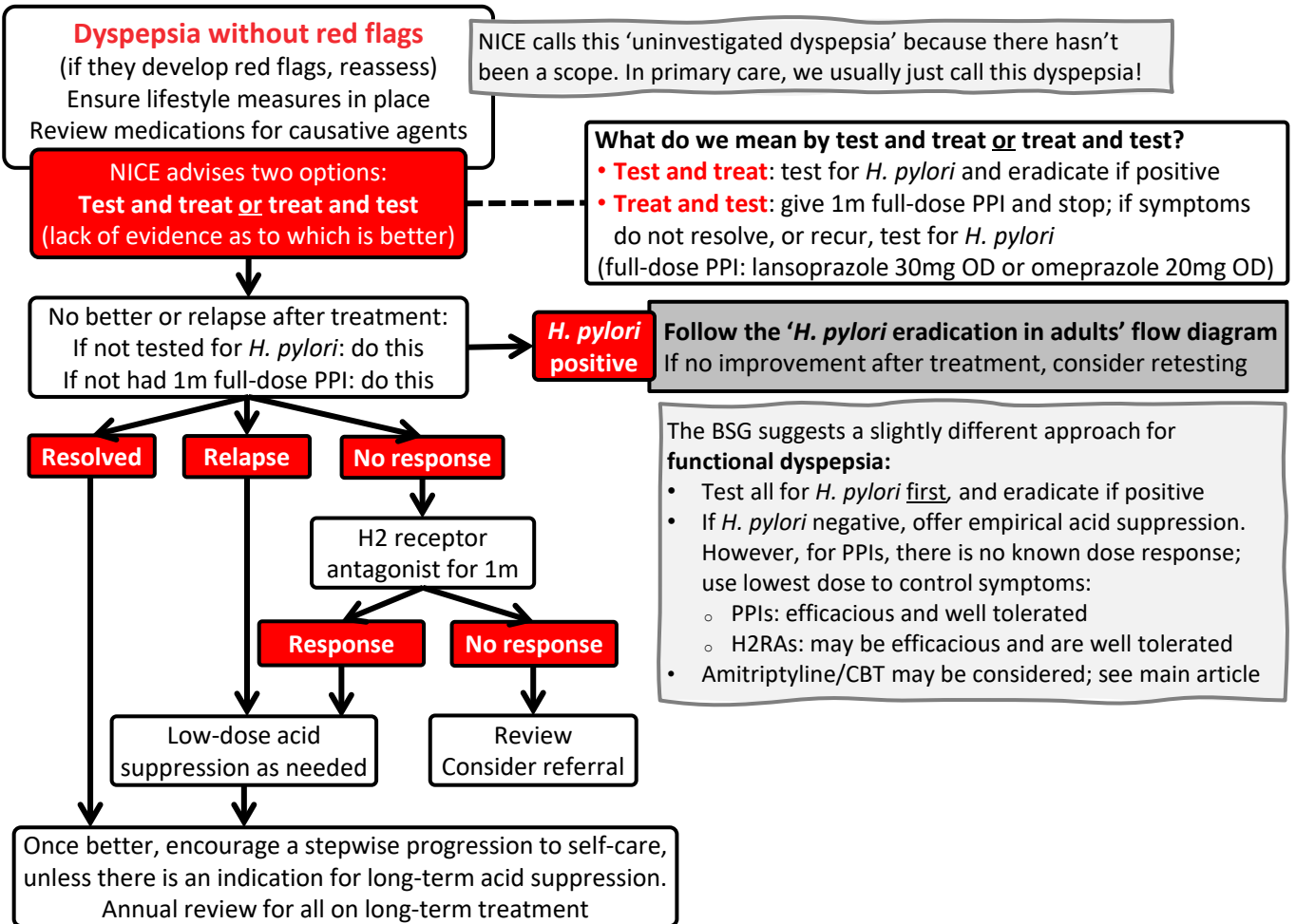
NICE 2019, CG184, Gut 2022;71:1697, Gut 2017;66:6, NEJM 2019;380:1158,  
PHE test and treat for *Helicobacter pylori* in dyspepsia, 2017



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## H. pylori testing

The prevalence of *H. pylori* has fallen in the past few decades, but it remains higher in certain groups. In these groups, consider a TEST and treat strategy rather than treat and test (PHE):

- Older people
- Those of North African ethnicity
- Those living in a known high-risk area
- Unexplained iron deficiency anaemia (after OGD and after other causes excluded)
- Previous gastric/duodenal ulcer or bleed AND not previously tested
- Previous gastric/duodenal ulcer or bleed: before taking NSAIDs

### H. pylori testing: practicalities

- Use either stool antigen or breath test (serology not recommended – specificity lower)
- Stop PPI for 2w (and antibiotics and bismuth for 4w) before testing:
  - If significant flare of symptoms on stopping PPI, can cover with a H2RA

### When to retest after H. pylori eradication?

- Severe, recurrent or persistent symptoms
- Persistent symptoms and the test was done within 2w of a PPI or 4w of antibiotics
- Poor compliance or high local resistance rates
- Duodenal ulcer, gastric ulcer, MALToma or after resection of early gastric cancer
- Increased risk of gastric cancer
- Taking aspirin without co-prescribed PPI (guidance doesn't explain why)

Retest in the absence of ongoing symptoms (test of cure)

### If retesting for H. pylori, wait 4–8w. Use H2RA cover if required

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# *H. pylori* eradication in adults

(NICE 2019, CG184, BNF accessed December 2022)



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Check for penicillin allergy

NOT allergic to penicillin

Clarithromycin OR metronidazole  
AND  
Amoxicillin  
AND  
Twice-daily PPI

**How to decide whether to use clarithromycin or metronidazole:**

- If previous use of clarithro/metronidazole, choose the other one
- If previously used BOTH, choose either
- No recent use: choose either!

Treatment fails

NO clarithromycin or metronidazole before started eradication

Clarithromycin OR metronidazole  
(use the one NOT used previously)  
AND  
Amoxicillin  
AND  
Twice-daily PPI

PREVIOUS clarithromycin or metronidazole before eradication, and used the other one for first-line eradication

Tetracycline (if CI, levofloxacin)  
AND  
Amoxicillin  
AND  
Twice-daily PPI

Treatment fails: gastroenterology advice

ALLERGIC to penicillin

Doses: all for 7d

## ANTIBIOTICS:

- Amoxicillin: 1g twice daily
- Clarithromycin: 500mg twice daily
- Metronidazole: 400mg twice daily
- Levofloxacin: 250mg twice daily
- Tetracycline: 500mg four times a day

## BISMUTH:

- Bismuth subsalicylate: 525mg four times a day

## PPIs:

- Omeprazole: 20-40mg twice daily
- Lansoprazole: 30mg twice daily
- Esomeprazole 20mg twice daily
- Pantoprazole: 40mg twice daily
- Rabeprazole: 20mg twice daily

(There are concerns around quinolone safety; see *Quinolones (the -floxacin): the risks* article)

## Why so complicated?

Because of concerns about antibiotic resistance. Also, *H. pylori* may have been present for some time so previous antibiotic exposure is important

Have they been prescribed clarithromycin before?

Not had clarithromycin

Clarithromycin  
AND  
Metronidazole  
AND  
Twice-daily PPI

Previous clarithromycin

Bismuth  
AND  
Metronidazole  
AND  
Tetracycline  
AND  
Twice-daily PPI

Treatment fails: previous exposure to quinolones?

Not had quinolones

Levofloxacin  
AND  
Metronidazole  
AND  
Twice-daily PPI

Previous quinolones

Bismuth  
AND  
Metronidazole  
AND  
Tetracycline  
AND  
Twice-daily PPI

Treatment fails: gastroenterology advice

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# PPI deprescribing in adults

(NICE NG231, 2023, BMJ 2022;379:e069211, Lancet 2017;390:490, NICE 2019, CG184, Can Fam Physician 2017;63(5):354, Gastroenterology 2022;162(4):1334, Gut 2022;71:1697, BJGP 2022 DOI: <https://doi.org/10.3399/bjgp22X721157>)



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For some people, long-term PPI use is clinically appropriate; for others, stopping treatment may be an option  
Who needs a long-term PPI and who can be considered for stepwise return to self-care?

If taken a PPI for >8w, assess whether long-term use is required

## If ANY of the following, continue PPI

- History of severe oesophagitis OR oesophageal ulcer OR peptic/oesophageal stricture
- Zollinger-Ellison syndrome
- High risk of GI bleeding, e.g.:
  - Chronic use of NSAIDs AND moderate/high bleeding risk (e.g. older, high NSAID dose, previous ulcer, renal disease, other drugs that also increase bleeding risk, e.g. steroids, antiplatelets)
  - Older people on aspirin (NNT to prevent one upper GI bleed over 5y is 25 if >85y)
- History of eosinophilic oesophagitis or idiopathic pulmonary fibrosis
- Patient not amenable to stopping PPI
- History of a gastric or duodenal ulcer:
  - NICE says once healed, low-dose maintenance or as-required PPI can be used
  - Canadian and US guidance says to continue PPI if a history of a bleeding GI ulcer

If PPIs needed long term, consider 'indication prescribing' so rationale is clear, e.g. *Take one daily for ZE syndrome*

Are long-term PPIs needed for Barrett's oesophagus? Yes, for symptom control. NICE does not recommend PPIs to prevent progression to cancer (insufficient evidence). Exception: if 'indefinite' dysplasia, need PPI+OGD every 6m

## If NONE of the above, consider deprescribing

- Is the patient amenable? (shared decision-making)
- Examples of clinical situations where deprescribing may be appropriate:
  - Mild-moderate oesophagitis and symptoms now controlled
  - Gastro-oesophageal reflux disease, had 4-8w PPI and symptoms now controlled
  - Functional dyspepsia or uninvestigated dyspepsia, had 2-12w PPI and symptoms now controlled
  - Treated for *H. pylori* and now asymptomatic
  - Started on stress ulcer prophylaxis in ITU and not stopped at discharge

## Benefits and risks of deprescribing

**Risks:** return of symptoms

**Benefits:** decreased pill burden, decreased drug side-effects/risks

**PPIs have been associated with the following** (these are associations, not causation): CVD, CKD, cancer (all cancers and GI cancers), fractures, dementia, hyponatraemia, hypomagnesaemia, B12 deficiency and death

(For more information on PPI safety concerns, see our article *Proton pump inhibitors: safety concerns*)

## Options for deprescribing

- **Taper the dose:** 1 in 10 will get return of symptoms (preferred option of US and Canadian guidance)
- **Use on demand/as required:** 1 in 10 will get return of symptoms (preferred option of Canadian guidance)
- **Change to H2RA:** about 1 in 5 will get return of symptoms (or consider over-the-counter treatment)
- **Stop abruptly:** about 1 in 2 will get return of symptoms (an option in US guidance)

## Troubleshooting rebound symptoms

- Rebound symptoms can occur; patient awareness is key. OTC antacids and lifestyle measures can help
- Symptoms are usually mild, manageable and settle within a few days to weeks
- Symptoms do not necessarily mean return of the underlying cause
- Make a plan for if symptoms are severe, frequent or persistent, e.g. restart PPI, reconsult

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